

<i>typevar</i> , X	type variable
<i>termvar</i> , x	term variable
<i>label</i> , l , k	field label
<i>index</i> , i , j , n , m	indices

T, S, U	$::=$		type
		X	type variable
		Top	maximum type
		$T \rightarrow T'$	type of functions
		$\forall X <: T. T'$	universal type
		$\{l_1 : T_1, \dots, l_n : T_n\}$	record
		(T)	S
		$[X \mapsto T] T'$	M
t	$::=$		term
		x	variable
		$\lambda x : T. t$	bind x in t abstraction
		$t t'$	application
		$\Lambda X <: T. t$	bind X in t type abstraction
		$t[T]$	type application
		$\{l_1 = t_1, \dots, l_n = t_n\}$	record
		$t.l$	projection
		let $p = t$ in t'	bind $b(p)$ in t' pattern binding
		(t)	S
		$[x \mapsto t] t'$	M
		$[X \mapsto T] t$	M
		σt	M
p	$::=$		pattern
		$x : T$	$b = x$ variable pattern
		$\{l_1 = p_1, \dots, l_n = p_n\}$	$b = b(p_1..p_n)$ record pattern
v	$::=$		values
		$\lambda x : T. t$	bind x in t abstraction
		$\Lambda X <: T. t$	bind X in t type abstraction
		$\{l_1 = v_1, \dots, l_n = v_n\}$	record
Γ, Δ	$::=$		type environment
		empty	
		$\Gamma, X <: T$	
		$\Gamma, x : T$	
		$\Gamma_1, \dots, \Gamma_n$	M
		$[X \mapsto T] \Gamma$	M
σ	$::=$		multiple term substitution
		$[x \mapsto t]$	
		$\sigma_1, \dots, \sigma_n$	
<i>terminals</i>	$::=$		
		λ	
		$\rightarrow$	
		$\Rightarrow$	
		$\vdash$	
		$\longrightarrow$	

	$\forall$ $<:$ $\mapsto$ $\wedge$ $\vee$ $=$	
<i>formula</i>	$::=$ $\mid$ <i>judgement</i> $\mid$ $x=x'$ $\mid$ $X=X'$ $\mid$ (<i>formula</i>) $\mid$ $\neg$ <i>formula</i> $\mid$ $\forall i \in 1..m.$ <i>formula</i> $\mid$ $\exists i \in 1..m.$ <i>formula</i> $\mid$ <i>formula</i> $\wedge$ <i>formula'</i> $\mid$ $l=l'$ $\mid$ <i>formula</i> ₁ ... <i>formula</i> _n	
<i>Judgement_in</i>	$::=$ $\mid$ $x \in \mathbf{dom}(\Gamma)$ $\mid$ $X \in \mathbf{dom}(\Gamma)$ $\mid$ $x : T \in \Gamma$ $\mid$ $X <: U \in \Gamma$	
<i>Jtype</i>	$::=$ $\mid$ $\Gamma \vdash \mathbf{ok}$ $\mid$ $\Gamma \vdash T$ $\mid$ $\Gamma \vdash S <: T$ $\mid$ $\Gamma \vdash k\ S <: f\{l_1 : T_1, \dots, l_n : T_n\}$ $\mid$ $\Gamma \vdash t : T$ $\mid$ $\vdash p : T \Rightarrow \Delta$	type environment Γ is well-formed type T is well-formed in type environment Γ S is a subtype of T field $k:S$ is matched, with subtypes $T_1, \dots, T_n$ term t has type T pattern p matches type T giving Δ
<i>Jop</i>	$::=$ $\mid$ $t_1 \longrightarrow t_2$ $\mid$ match (p, v)= σ $\mid$ matchfield (l, p) vs $\{k_1=v_1, \dots, k_n=v_n\}$ gives σ	t_1 reduces to t_2
<i>judgement</i>	$::=$ $\mid$ <i>Judgement_in</i> $\mid$ <i>Jtype</i> $\mid$ <i>Jop</i>	
<i>user_syntax</i>	$::=$ $\mid$ <i>typevar</i> $\mid$ <i>termvar</i> $\mid$ <i>label</i> $\mid$ <i>index</i> $\mid$ T	

	t
	p
	v
	Γ
	σ
	<i>terminals</i>
	<i>formula</i>

$$\boxed{x \in \mathbf{dom}(\Gamma)}$$

$$\frac{}{x \in \mathbf{dom}(\Gamma, x : T)} \quad \text{XING_1}$$

$$\frac{x \in \mathbf{dom}(\Gamma)}{x \in \mathbf{dom}(\Gamma, X' <: U')} \quad \text{XING_2}$$

$$\frac{x \in \mathbf{dom}(\Gamma)}{x \in \mathbf{dom}(\Gamma, x' : T')} \quad \text{XING_3}$$

$$\boxed{X \in \mathbf{dom}(\Gamma)}$$

$$\frac{}{X \in \mathbf{dom}(\Gamma, X <: U)} \quad \text{XING_1}$$

$$\frac{X \in \mathbf{dom}(\Gamma)}{X \in \mathbf{dom}(\Gamma, X' <: U')} \quad \text{XING_2}$$

$$\frac{X \in \mathbf{dom}(\Gamma)}{X \in \mathbf{dom}(\Gamma, x' : T')} \quad \text{XING_3}$$

$$\boxed{x : T \in \Gamma}$$

$$\frac{}{x : T \in \Gamma, x : T} \quad \text{TIN_1}$$

$$\frac{x : T \in \Gamma}{x : T \in \Gamma, X' <: U'} \quad \text{TIN_2}$$

$$\frac{x : T \in \Gamma}{x : T \in \Gamma, x' : T'} \quad \text{TIN_3}$$

$$\boxed{X <: U \in \Gamma}$$

$$\frac{}{X <: U \in \Gamma, X <: U} \quad \text{TIN_1}$$

$$\frac{X <: U \in \Gamma}{X <: U \in \Gamma, X' <: U'} \quad \text{TIN_2}$$

$$\frac{X <: U \in \Gamma}{X <: U \in \Gamma, x' : T'} \quad \text{TIN_3}$$

$$\boxed{\Gamma \vdash \mathbf{ok}} \quad \text{type environment } \Gamma \text{ is well-formed}$$

$$\frac{}{\mathbf{empty} \vdash \mathbf{ok}} \quad \text{GOK_1}$$

$$\frac{\Gamma \vdash T \quad \neg(x \in \mathbf{dom}(\Gamma))}{\Gamma, x : T \vdash \mathbf{ok}} \quad \text{GOK_2}$$

$$\frac{\Gamma \vdash T \quad \neg(X \in \mathbf{dom}(\Gamma))}{\Gamma, X <: T \vdash \mathbf{ok}} \quad \text{GOK_3}$$

$\boxed{\Gamma \vdash T}$ type T is well-formed in type environment Γ

$$\frac{\Gamma \vdash \mathbf{ok} \quad X <: U \in \Gamma}{\Gamma \vdash X} \quad \text{GT_VAR}$$

$$\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash \mathbf{Top}} \quad \text{GT_TOP}$$

$$\frac{\Gamma \vdash T \quad \Gamma \vdash T'}{\Gamma \vdash T \rightarrow T'} \quad \text{GT_FUN}$$

$$\frac{\Gamma, X <: T \vdash T'}{\Gamma \vdash \forall X <: T. T'} \quad \text{GT_FORALL}$$

$$\frac{\Gamma \vdash T_1 \quad \dots \quad \Gamma \vdash T_n}{\Gamma \vdash \{l_1 : T_1, \dots, l_n : T_n\}} \quad \text{GT_RCD}$$

$\boxed{\Gamma \vdash S <: T}$ S is a subtype of T

$$\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash S <: \mathbf{Top}} \quad \text{SA_TOP}$$

$$\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash X <: X} \quad \text{SA_REFL_TVAR}$$

$$\frac{X <: U \in \Gamma \quad \Gamma \vdash U <: T}{\Gamma \vdash X <: T} \quad \text{SA_TRANS_TVAR}$$

$$\frac{\Gamma \vdash T_1 <: S_1 \quad \Gamma \vdash S_2 <: T_2}{\Gamma \vdash S_1 \rightarrow S_2 <: T_1 \rightarrow T_2} \quad \text{SA_ARROW}$$

$$\frac{\Gamma \vdash T_1 <: S_1 \quad \Gamma, X <: T_1 \vdash S_2 <: T_2}{\Gamma \vdash \forall X <: S_1. S_2 <: \forall X <: T_1. T_2} \quad \text{SA_ALL}$$

$$\frac{\forall i \in 1..m. \Gamma \vdash k_i S_i <: f\{l_1 : T_1, \dots, l_n : T_n\}}{\Gamma \vdash \{k_1 : S_1, \dots, k_m : S_m\} <: \{l_1 : T_1, \dots, l_n : T_n\}} \quad \text{SA_RCD}$$

$\boxed{\Gamma \vdash k S <: f\{l_1 : T_1, \dots, l_n : T_n\}}$ field $k:S$ is matched, with subtyping, by a field of the record

$$\frac{\Gamma \vdash S <: T}{\Gamma \vdash k S <: f\{k : T, l_1 : T_1, \dots, l_n : T_n\}} \quad \text{SAF_FOUND}$$

$$\frac{\Gamma \vdash k S <: f\{l_1 : T_1, \dots, l_n : T_n\}}{\Gamma \vdash k S <: f\{l : T, l_1 : T_1, \dots, l_n : T_n\}} \quad \text{SAF_SKIP}$$

$\boxed{\Gamma \vdash t : T}$ term t has type T

$$\frac{\Gamma \vdash \mathbf{ok} \quad x : T \in \Gamma}{\Gamma \vdash x : T} \quad \text{TYCH_VAR}$$

$$\begin{array}{c}
\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x:T_1. t_2 : T_1 \rightarrow T_2} \quad \text{TYCH_ABS} \\
\\
\frac{\Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \quad \Gamma \vdash t_2 : T_{11}}{\Gamma \vdash t_1 t_2 : T_{12}} \quad \text{TYCH_APP} \\
\\
\frac{\Gamma, X <: T_1 \vdash t_2 : T_2}{\Gamma \vdash \Lambda X <: T_1. t_2 : \forall X <: T_1. T_2} \quad \text{TYCH_TABS} \\
\\
\frac{\Gamma \vdash t_1 : \forall X <: T_{11}. T_{12} \quad \Gamma \vdash T_2 <: T_{11}}{\Gamma \vdash t_1[T_2] : [X \mapsto T_2] T_{12}} \quad \text{TYCH_TAPP} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \quad \vdash p : T_1 \Rightarrow \Delta \quad \Gamma, \Delta \vdash t_2 : T_2}{\Gamma \vdash \text{let } p=t_1 \text{ in } t_2 : T_2} \quad \text{TYCH_LET} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \quad \dots \quad \Gamma \vdash t_n : T_n}{\Gamma \vdash \{l_1=t_1, \dots, l_n=t_n\} : \{l_1 : T_1, \dots, l_n : T_n\}} \quad \text{TYCH_RCD} \\
\\
\frac{\Gamma \vdash t : \{l_1 : T_1, \dots, l_n : T_n\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TYCH_PROJ} \\
\\
\frac{\Gamma \vdash t : S \quad \Gamma \vdash S <: T}{\Gamma \vdash t : T} \quad \text{TYCH_SUB}
\end{array}$$

$\boxed{\vdash p : T \Rightarrow \Delta}$ pattern p matches type T giving bindings Δ

$$\begin{array}{c}
\frac{}{\vdash x : T : T \Rightarrow \text{empty}, x : T} \quad \text{PAT_VAR} \\
\\
\frac{\vdash p_1 : T_1 \Rightarrow \Delta_1 \quad \dots \quad \vdash p_n : T_n \Rightarrow \Delta_n}{\vdash \{l_1=p_1, \dots, l_n=p_n\} : \{l_1 : T_1, \dots, l_n : T_n\} \Rightarrow \Delta_1, \dots, \Delta_n} \quad \text{PAT_RCD}
\end{array}$$

$\boxed{t_1 \longrightarrow t_2}$ t_1 reduces to t_2

$$\begin{array}{c}
\frac{}{(\lambda x:T_{11}. t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \quad \text{REDUCE_APPABS} \\
\\
\frac{}{(\Lambda X <: T_{11}. t_{12})[T_2] \longrightarrow [X \mapsto T_2] t_{12}} \quad \text{REDUCE_TAPPTABS} \\
\\
\frac{\text{match}(p, v_1) = \sigma}{\text{let } p=v_1 \text{ in } t_2 \longrightarrow \sigma t_2} \quad \text{REDUCE_LETV} \\
\\
\frac{}{\{l'_1=v_1, \dots, l'_n=v_n\}.l'_j \longrightarrow v_j} \quad \text{REDUCE_PROJRCD} \\
\\
\frac{t_1 \longrightarrow t'_1}{t_1 t \longrightarrow t'_1 t} \quad \text{REDUCE_CTX_APP_FUN} \\
\\
\frac{t_1 \longrightarrow t'_1}{v t_1 \longrightarrow v t'_1} \quad \text{REDUCE_CTX_APP_ARG} \\
\\
\frac{t_1 \longrightarrow t'_1}{t_1[T] \longrightarrow t'_1[T]} \quad \text{REDUCE_CTX_TYPE_FUN}
\end{array}$$

$$\frac{t \longrightarrow t'}{\{l_1=v_1, \dots, l_m=v_m, l=t, l'_1=t'_1, \dots, l'_n=t'_n\} \longrightarrow \{l_1=v_1, \dots, l_m=v_m, l=t', l'_1=t'_1, \dots, l'_n=t'_n\}} \quad \text{REDUCE_CTX_RECORD}$$

$$\frac{t_1 \longrightarrow t'_1}{\text{let } p=t_1 \text{ in } t_2 \longrightarrow \text{let } p=t'_1 \text{ in } t_2} \quad \text{REDUCE_CTX_LET_BINDING}$$

$$\boxed{\text{match}(p, v)=\sigma}$$

$$\frac{}{\text{match}(x : T, v)=[x \mapsto v]} \quad \text{M_VAR}$$

$$\frac{\forall i \in 1..m. \text{matchfield}(l_i, p_i) \text{ vs } \{k_1=v_1, \dots, k_n=v_n\} \text{ gives } \sigma_i}{\text{match}(\{l_1=p_1, \dots, l_m=p_m\}, \{k_1=v_1, \dots, k_n=v_n\})=\sigma_1, \dots, \sigma_m} \quad \text{M_RCD}$$

$$\boxed{\text{matchfield}(l, p) \text{ vs } \{k_1=v_1, \dots, k_n=v_n\} \text{ gives } \sigma}$$

$$\frac{\text{match}(p, v)=\sigma}{\text{matchfield}(l, p) \text{ vs } \{l=v, k_1=v_1, \dots, k_n=v_n\} \text{ gives } \sigma} \quad \text{MF_FOUND}$$

$$\frac{\text{matchfield}(l, p) \text{ vs } \{k_1=v_1, \dots, k_n=v_n\} \text{ gives } \sigma}{\text{matchfield}(l, p) \text{ vs } \{k=v, k_1=v_1, \dots, k_n=v_n\} \text{ gives } \sigma} \quad \text{MF_SKIP}$$

Definition rules: 52 good 0 bad

Definition rule clauses: 107 good 0 bad